

A SECTION OF RESEARCH STATEMENT (DEC 17, 2017)

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0.1. Smooth proper schemes over $\mathbb{Z}$; good and bad reduction of varieties over $\mathbb{Q}$. A folklore conjecture says that a smooth proper scheme X over $\mathbb{Z}$ is geometrically rational [1]. In relative dimension 1, the only such scheme is $\mathbb{P}_{\mathbb{Z}}^1$, by the non-existence of nontrivial abelian scheme over $\mathbb{Z}$ proved independently by Abrashkin and Fontaine by studying the Galois actions on étale cohomology groups and finite flat group schemes and using Odlyzko's discriminant bound to conclude the argument. Later, they independently showed that the Betti and Hodge numbers of $X_{\mathbb{C}}$ satisfy $h^2(X_{\mathbb{C}}) = h^{1,1}(X_{\mathbb{C}})$ and $h^1(X_{\mathbb{C}}) = h^3(X_{\mathbb{C}}) = 0$ using p -adic Hodge theory, and Abrashkin showed under GRH that $h^4(X_{\mathbb{C}}) = h^{2,2}(X_{\mathbb{C}})$ (see [2]). Therefore in relative dimension 2, Enriques surfaces become the first natural examples to consider, as they are neither geometrically rational nor ruled out by these results. Another interesting class of examples to consider is quiver varieties (moduli spaces of certain representations of quivers with fixed dimension vectors). For many surfaces possessing a full strong exceptional collection, it is known that one can recover the surface as the quiver variety of the quiver of sections of the exceptional collection, if a suitable stability condition is imposed on the representations. I am currently looking for possible connections between primes of bad reduction of the surface and modular representation theory of the quiver (with relations).

The result of Abrashkin and Fontaine also holds if we start with a projective variety $X/\mathbb{Q}$ that has good reduction everywhere. An Enriques surface admits a canonical double cover that is a K3 surface. If an Enriques surface over $\mathbb{Q}$ has good reduction everywhere, its K3 double cover has good reduction except at 2. It is expected that such K3 surfaces over $\mathbb{Q}$ are finite in number [3]; André [4] almost proved it, except that he considered polarized good reduction only. It is desirable to list all these K3 surfaces, which hopefully can be done by applying the Kuga–Satake construction to reduce it to a problem about abelian varieties with bad reduction only at 2, about which there are known results.

A related question is: when can a smooth complete varieties over $\mathbb{Q}$ be realized as the generic fiber of a smooth proper scheme over $\mathbb{Z}$, i.e. have a smooth proper model over $\mathbb{Z}$? Such a variety must have good reduction everywhere, and the converse holds for relative curves and abelian schemes, and it is interesting to investigate in what other cases it holds. Good reduction implies in turn that the Galois actions on the p -adic étale cohomology groups are unramified at $\ell \neq p$ and crystalline at p , but not vice versa. For example, Brauer–Severi varieties have unramified/crystalline cohomology, but do not have good reduction except for $\mathbb{P}_{\mathbb{Q}}^n$ [5]. An example of an Enriques surface over $\mathbb{Z}[1/2]$ with unramified cohomology has also been constructed, and it is not known whether it has good reduction at 2 [6]. The recent work [7] of Bhatt, Morrow and Scholze on integral p -adic Hodge theory contains an explanation of the pathologies in the cohomology of Enriques surfaces in characteristic 2, and their work may prove useful for our purpose. Also, rational varieties have Tate type cohomology, which is automatically unramified/crystalline, so it seems challenging to show that they may have bad reduction. It would be desirable to come up with general obstructions to good reduction and existence of a smooth proper model that apply to such cases. It is also natural to consider the transition between unramified/crystalline cohomology, everywhere good reduction, and smooth proper models, over other number rings.

Any even unimodular lattice of rank n gives rise to a smooth quadric X in $\mathbb{P}_{\mathbb{Z}}^{n-1}$ of relative dimension $n - 2$ over $\mathbb{Z}$ [8], and some other lattices, for example the E_7 lattice, does the same [9]. If the lattice is positive definite, X has no $\mathbb{Q}$ -points and hence is not rational (though it is still geometrically rational). Therefore, if d is the minimal relative dimension of a proper smooth scheme over $\mathbb{Z}$ that is not rational, we have $2 \leq d \leq 5$, and it is an interesting problem to determine the exact value of d . So far I am able to show that no such quadric with $d < 5$ exists.

REFERENCES

- [1] B. Wieland, Does smooth and proper over $\mathbb{Z}$ imply rational? <https://mathoverflow.net/questions/97086/>.
- [2] V. Abrashkin, A semi-stable case of the Shafarevich Conjecture, <http://www.maths.dur.ac.uk/~dma0va/DurLMS.pdf>.
- [3] J. S. Ellenberg, K3 surfaces with good reduction away from finitely many places, <https://mathoverflow.net/questions/4695/>.
- [4] Y. André, On the Shafarevich and Tate conjectures for hyper-Kähler varieties, Math. Ann. 305 (1996), no. 2, 205–248.

- [5] C. S. Dalawat, Good reduction, bad reduction, arXiv:0605326.
- [6] W. Sawin, Enriques surfaces over $\mathbb{Z}$, <https://mathoverflow.net/questions/201273/>.
- [7] B. Bhatt, M. Morrow and P. Scholze, Integral p -adic Hodge theory, arXiv:1602.03148.
- [8] K. Buzzard, Smooth proper scheme over $\mathbb{Z}$, <https://mathoverflow.net/a/9605/3332>.
- [9] W. Sawin, What can be the dimension of a pointless smooth proper $\mathbb{Z}$ -scheme? <https://mathoverflow.net/q/111708/>.